NITISH PRIYAN R S

Computer Science & Engineering Student | Software Engineer | Java Developer | AI/ML Engineer
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PROFESSIONAL SUMMARY

Computer Science undergraduate (Batch of 2028) and aspiring Software Engineer with strong proficiency in Java, backed by an Oracle Certified Associate Java SE 17 Developer credential and AWS Cloud Practitioner certification. Hands-on experience across Full-Stack Development, Machine Learning and MLOps. Built and deployed 10+ projects spanning a patent-filed AI e-governance platform, an AI campus digital twin, and MLOps decision-support systems using Java, Python, TypeScript, REST APIs, React, and Git. Solved 250+ DSA problems in Java on LeetCode, with strong foundations in Object-Oriented Programming.

TECHNICAL SKILLS

Programming Languages: Java (Oracle Certified, SE 17), Python, TypeScript, C, C++, SQL
Machine Learning & AI: Scikit-learn, XGBoost, MLflow, Gemini AI
Web & Backend Development: FastAPI, Node.js, React, Three.js, REST API Design, JWT Authentication
Databases & DevOps: MySQL, PostgreSQL, MongoDB, SQLAlchemy, Docker, Git, GitHub
Core Concepts: Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), MLOps, API Integration, Problem Solving

PROJECT EXPERIENCE

GovScheme — AI-Powered Welfare Discovery & Lifecycle Administration Platform (Patent Filed) | GitHub

(Individual)

React, TypeScript, Node.js, MongoDB, Google Gemini AI, WebSockets, Tesseract.js

- Architected and patented an enterprise multilingual e-governance platform that empowers citizens across 10 official Indian languages to discover, evaluate, and track 75+ welfare schemes.
- Engineered a proprietary hybrid decision engine combining deterministic constraint checking with Multi-Criteria Decision Analysis (MCDA) and localized semantic search to eliminate LLM hallucinations and achieve sub-50ms explainable eligibility matching.
- Integrated a privacy-preserving client-side OCR pipeline using Tesseract.js worker threads for automated document verification, and a streaming bidirectional Indic voice interface (Web Speech API + Neural TTS).
- Built a real-time WebSocket-powered application tracker featuring multi-tier provenance logging and direct 1-click redirection to official .gov.in portals.

NEXORA — AI-Powered Campus Digital Twin

Python, FastAPI, React.js, Three.js, Scikit-learn, SQLAlchemy, PostgreSQL, SQLite, Docker, JWT, Git

- Developed an AI-powered campus digital twin platform to monitor and analyze simulated energy, water, occupancy, parking, and network systems.
- Implemented predictive analytics using five Scikit-learn models for campus resource forecasting and anomaly detection, alongside what-if simulations, optimization workflows, AI decision support, and sustainability analytics.
- Built an interactive 3D campus visualization with React, Three.js, and React Three Fiber, backed by a FastAPI service with JWT authentication, refresh-token rotation, and role-based access control.
- Achieved 129/129 passing automated tests; cut the initial frontend JS bundle from 1.73 MB to 253.33 kB.

Student Performance Prediction & MLOps Decision Support System | GitHub

Python, FastAPI, XGBoost, MLflow, DVC, Docker, CI/CD

- Built an end-to-end MLOps platform for student outcome prediction, risk scoring, and personalized intervention recommendations, achieving 89.1% prediction accuracy and 82.6% at-risk recall.
- Integrated MLflow for experiment tracking, DVC for data versioning, and Evidently AI for model/data drift monitoring within an automated CI/CD pipeline.
- Containerized the full pipeline with Docker and implemented asynchronous telemetry for real-time monitoring of model performance in production.

EDUCATION

Bannari Amman Institute of Technology — B.E. Computer Science and Engineering — Expected May 2028 | CGPA: 7.8
Saratha Matriculation Higher Secondary School — HSC — 2023 – 2024 | Score: 93.1%

ACHIEVEMENTS & ACTIVITIES

- Solved 250+ Data Structures and Algorithms (DSA) problems in Java on LeetCode.
- Selected participant, Smart India Hackathon (SIH) 2025 — national-level innovation and problem-solving competition.
- Patent filed for GovScheme's hybrid eligibility decision engine and e-governance platform.

CERTIFICATIONS

- Oracle Certified Associate: Java SE 17 Developer
- AWS Cloud Practitioner Essentials — Amazon Web Services (AWS)
- Introduction to Cybersecurity — Cisco Networking Academy

SOFT SKILLS & LANGUAGES

Teamwork: Collaborated cross-functionally on team-based software and ML projects
Other Soft Skills: Problem Solving, Critical Thinking, Communication
Languages Known: English (Professional Proficiency), Tamil (Native)